

CANDIDATE NAME _____ ()

CG _____



SERANGOON JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION 2018

H2 BIOLOGY

Paper 4 Practical

9744/04**2 hours 30 minutes**

READ THESE INSTRUCTIONS FIRST

Write your name, index number and CG in the spaces at the top of this page.

Give details of the practical shift and laboratory, where appropriate, in the boxes provided.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate. You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

Shift	
Laboratory	
For examiner's use	
1	/25
2	/17
3	/13
Total	/55

This question paper consists of **22** printed pages including this cover page.

Question 1

Yeast cells contain enzymes which catalyse the breakdown of glucose to produce carbon dioxide and water.

The carbon dioxide reacts with water and forms a weak acid.

Bromothymol blue is a pH indicator and changes colour as shown in Table 1.1.

Table 1.1

pH	colour of bromothymol blue
8	blue
7	green
6	yellow

You are required

- to immobilise the yeast cells in sodium alginate beads
- to follow a student's procedure to investigate the independent variable, changing the surface area of the beads.

You are provided with

- 15 cm³ of yeast suspension, labelled **Y**
 - 15 cm³ of 2.0% sodium alginate solution, labelled **SA**
 - 50 cm³ of 1.5% calcium chloride solution, labelled **C**
 - 40 cm³ of 2.0% glucose solution, labelled **G**
 - 50 cm³ of bromothymol blue, labelled **B**
 - 20 cm³ of sodium hydroxide solution, labelled **A**
1. Put 20 cm³ of **C** into a large test-tube.
 2. Put 5 cm³ of **SA** into a small beaker or container.
 3. Collect 5 cm³ of **Y** from below the froth and put it into the same container as **SA**. Mix well.
 4. Use a 5 cm³ syringe to collect 5 cm³ of the mixture **SA** and **Y**.

5. Suspend the 5 cm³ syringe over the large test-tube containing **C** as shown in Fig. 1.1.

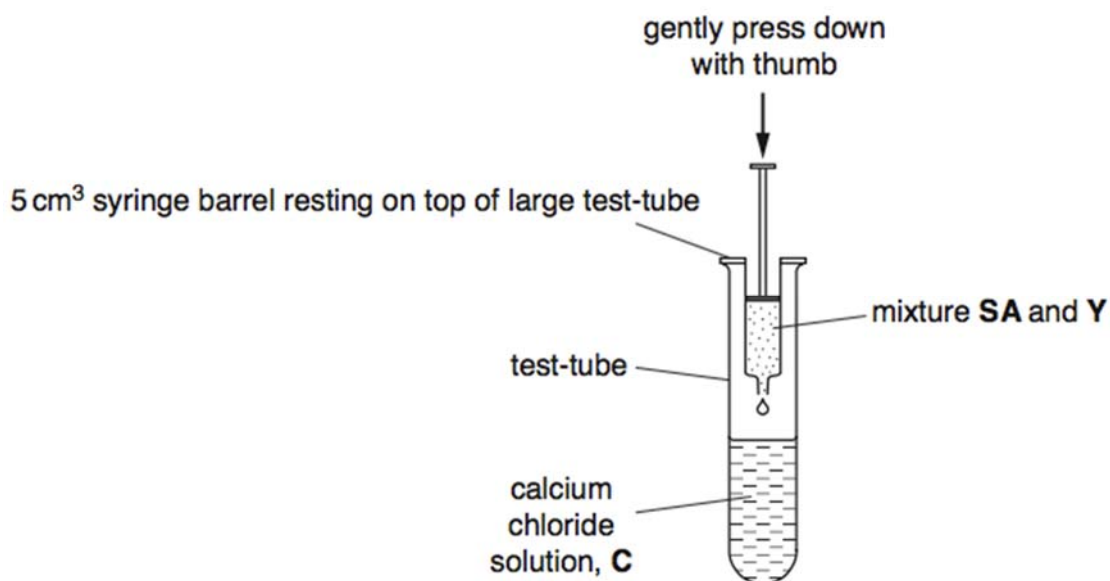


Fig. 1.1

6. Gently press down on the plunger of the syringe with your thumb to release a drop into solution **C**. The drop should form a bead.
7. Repeat step 6 to make the number of beads that you think you will need.
8. Tip the contents of the large test-tube into a Petri dish or shallow container.

You will need to calculate the mean surface area of the beads. Use a spatula or blunt forceps to pick up the beads.

To do this

- decide on the number of beads you will measure
- use the 2mm × 2mm grid to measure each bead
- calculate the surface area of each bead using the formula

$$\text{surface area} = 4\pi r^2 \text{ where } \pi = 3.14, r = \text{radius of a bead}$$

- calculate the mean surface area of the beads.

(a) (i) Prepare the space below to show your measurements and calculations.

1. Radius/ Diameter of at least 5 beads measured [MMO]
2. Records diameter/ radius with correct units, i.e. mm [MMO]
 - Reject: any measurements not whole number of mm
 - Reject: 6 mm or more for diameter or 3 mm or more for radius
3. One correct calculation for one surface area [ACE]
4. Shows addition of bead measurements divided by number measured OR each surface area added together and divided by number [PDO]
5. Answer no more than 3 sig. figs [PDO]

Show all the steps in your calculation of the mean.

mean surface area of the beadsmm² [5]

A student suggested that it was possible to investigate the independent variable, surface area, by changing the number of beads. The maximum number of beads used was 20. Decide the other numbers of beads to use and state the different number of beads you will use.

e.g. 5, 10, 15, 20 beads (*no credit awarded here*)

(ii) The student's procedure is shown below. State an appropriate hypothesis.

1. The greater the surface area of the alginate beads, the shorter the time taken for the solution to change from blue to green/ yellow.

[1]

Carry out the student's procedure.

9. Label as many small test-tubes as you will need with the number of beads for each test-tube.
10. Put 10 cm³ of solution **G** into each test-tube.
11. Put 1 cm³ of **B** into each test-tube. Put the bung in each test-tube in turn and mix.
12. If the contents of the test-tube are not blue, add one drop at a time of **A** to the contents of each test-tube to turn them all the same blue colour.
13. Put the required number of beads into each test-tube.
14. Put the bung in each test-tube in turn and mix contents. Mix every 2 minutes for 6 minutes.
15. Record your observations after each 2 minutes, up to 6 minutes.

(ii) Prepare the space below to record your observations.

1. Table with all cells drawn AND top/left heading – surface area/ mm² OR number of beads (no units) [PDO]
2. Top/ left Heading – colour/ observation at 2, 4 and 6 minutes [PDO]
3. Only records at 2, 4 and 6 minutes [MMO]
4. Highest number of beads (i.e. 20 beads) shows yellow/ green at 2 min [MMO]
5. Surface area recorded in table [MMO]
6. Use 20 beads in one tube and at least 3 different number of beads [MMO]
7. Even range [MMO]

[7]

- (iii) The student realised that there were two independent variables in this procedure.

State the **two** independent variables.

1. Surface area or number of beads AND enzyme or yeast concentration/ quantity
[ACE]
 - o Reject: more than two variables

[1]

- (iv) Suggest how you would make **three** improvements to the student's procedure.

1. Idea of cubes of different surface areas but with equal volume (Reject: amount) of yeast
2. Idea of equal shaking

[max 2 for the following]

3. Repeat measurements AND find mean/ average
4. Use of colorimeter/ white tile or pH paper or meter
5. Separate beads using a container of larger surface area e.g. Petri dish
6. Use of thermostatically controlled water bath
7. Idea of keeping time the same e.g. staggered start or have separate experiments
8. Use more beads or more surface areas

[ACE]

[3]

Question 1 continues on page 8

Yeast cells also contain catalase, which catalyses the breakdown of hydrogen peroxide into oxygen and water.

Another student investigated the evolution of oxygen during the breakdown of hydrogen peroxide. Immediately after some yeast immobilised in calcium alginate beads and the hydrogen peroxide were mixed, a stop clock was started and the volume of oxygen released in each minute for five minutes was recorded.

The student's results are shown in Table 1.2.

(b) (i) Complete Table 1.2.

Table 1.2

time / min	volume of oxygen collected in each minute / cm ³					
	trial 1	trial 2	trial 3	trial 4	trial 5	mean
1	3.0	2.8	3.0	2.9	2.8	2.90
2	0.6	0.8	0.8	0.7	0.9	0.76
3	0.4	0.5	0.6	0.6	0.7	0.56
4	0.3	0.3	0.4	0.5	0.5	0.40
5	0.1	0.2	0.2	0.1	0.3	0.18

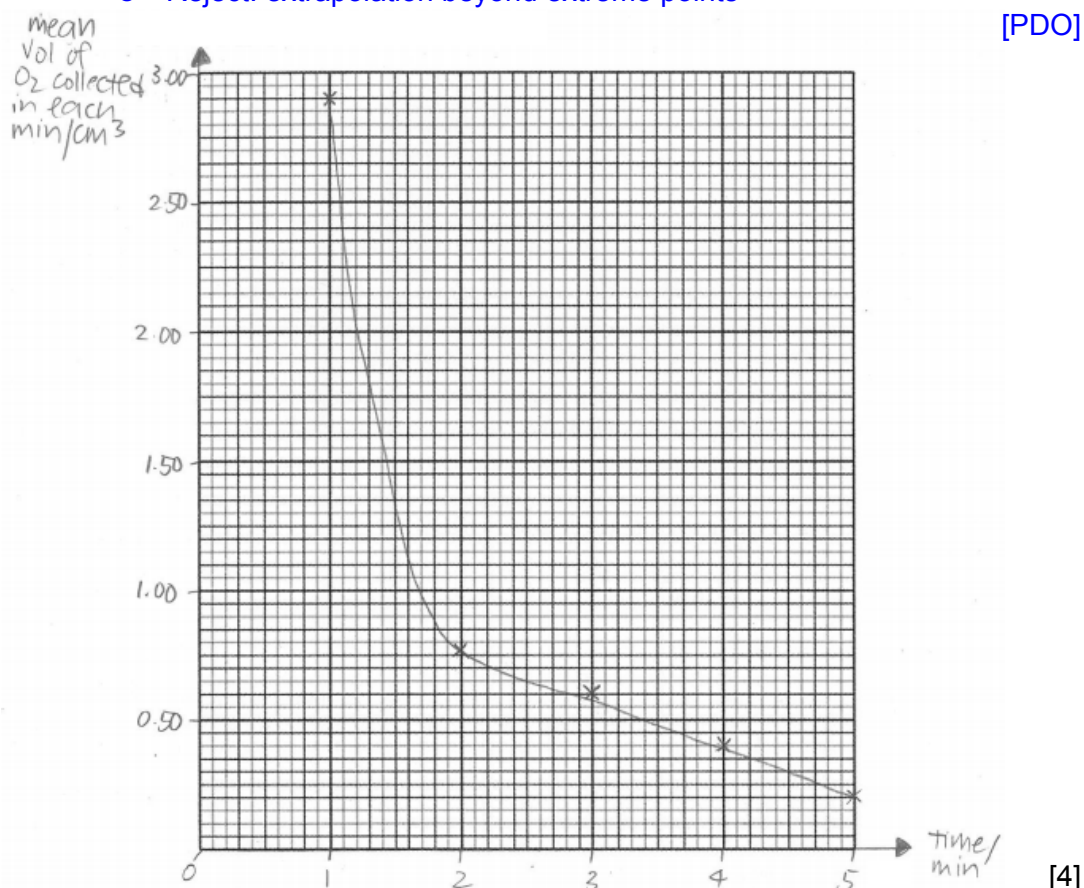
Answers to be to 2 dp (1 more than given data)

[1]

(ii) Plot a graph of the data in Table 1.2.

1. Orientation: x-axis – time/ min, y-axis - mean volume of oxygen collected/ cm^3
(Reject: missing units)
2. Scale: 10 small squares per 0.50 cm^3 (y-axis), 10 small squares per 1 minute (x axis). (Reject: awkward scale)
3. Plot: Correct plotting with crosses or dot in circle; intersection of cross must be clear to show plot. (Reject plotting if awkward scale)
4. Line: ruled/ straight line to all points or smooth curve.
 - Either ruled lines plot to plot, or curved line through all plots
 - Quality – no thicker than grid, not feathery
 - Reject: extrapolation beyond extreme points

[PDO]



(iii) Describe and explain the results of the student's investigation.

1. * Between the first to second minute, (idea of) volume of oxygen collected drops drastically from 2.90 to 0.76 cm^3
2. Idea of high rate of enzyme-substrate complex formation or high rate of hydrogen peroxide/ substrate fitting into or binding to active sites/ enzymes/ catalases
3. * From 2 to 5 min, volume of oxygen collected decreases gently/ less drastically from 0.76 to 0.18 cm^3
4. (Idea of) the lack of hydrogen peroxide/ substrate or hydrogen peroxide concentration not high enough

Max 3, * are compulsory points

[ACE]

[3]

[Total: 25]

Question 2

You are required to use a sharp pencil for drawings.

Iodine solution and methylene blue solution are used as stains for biological material.

You are required to:

- observe the effect of using the different stains, iodine solution, **I**, and methylene blue solution, **M**, on thin sections of onion tissue, **S1**
- observe and record the cells and their cell contents.

Iodine solution and **methylene blue** solution will stain your skin.

If any iodine solution or methylene blue solution comes into contact with your skin wash off immediately with water.

You are provided with:

- three pieces of onion tissue, in a dish labelled **S1**
- iodine solution, **I**
- methylene blue solution, **M**
- distilled water, **DW**.

You are required to:

- prepare **three** microscope slides of onion tissue, one using iodine solution, **I**, one using methylene blue solution, **M**, and one using distilled water, **DW**
- use the microscope to observe the onion cells after **I**, **M** and **DW** have been added
- record your observations by using annotated drawings of **two** adjacent onion cells from each of the prepared slides.

Proceed as follows:

1. Label three dry and clean microscope slides, **I**, **M** and **DW** and put the slides on a paper towel.
2. Put a few drops of:
 - iodine solution onto slide **I**
 - methylene blue solution onto slide **M**
 - distilled water onto slide **DW**.

3. Remove a piece of the onion tissue from **S1** and, using forceps or fingers, peel off the inner concave epidermis as shown in Fig. 2.1.

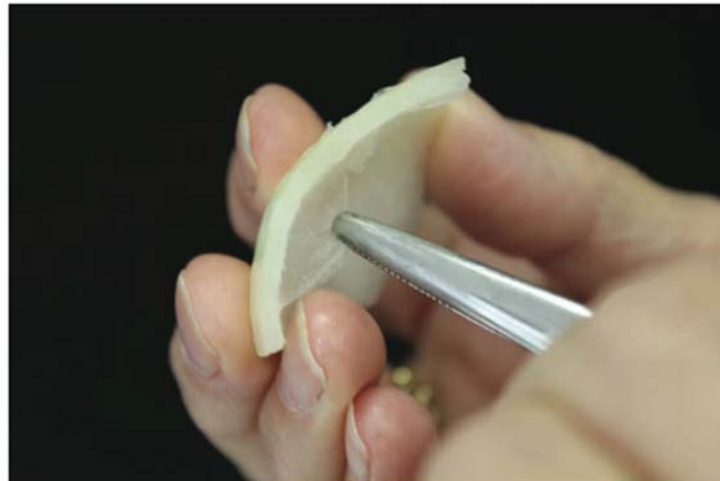


Fig. 2.1

4. Cut three pieces of the epidermis, each smaller than a coverslip.
5. Place one piece of the epidermis onto each of the slides as shown as Fig. 2.2. If the epidermis is folded, you may need to add more drops of **I** or **M** or **DW** so that it floats or uncurls.

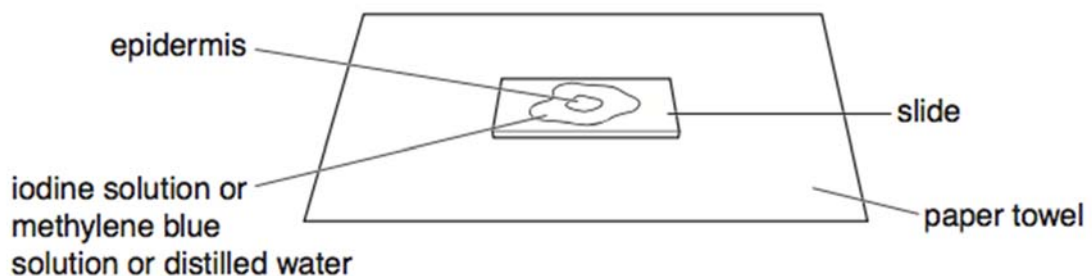


Fig. 2.2

6. Cover the epidermis on the slide with a coverslip and use a paper towel to remove any excess liquid that is outside the coverslip.
7. View the slide using the microscope. Look for the thinnest part of the section so that the cells and their contents can be observed.

You may need to reduce the amount of light entering the microscope to observe the cells.

(a) (i) Make a large drawing of **two** adjacent cells with any observable cell contents from each of:

- slide **I**
- slide **M**
- slide **DW**.

Use **one** ruled label line and label to show **one** nucleus on **one** of your drawings.

*cells from slide **I***

1. two adjacent cells each for **I**, **M** and **DW**
2. sharp continuous lines for all cells + widest cell is at least 60 mm across its widest point
3. cells drawn to correct shape (elongated + angular edges)
4. double lines for cell walls (drawn to correct proportion) + middle lamella
5. at least one nucleus drawn (to correct proportion)
6. label with label line to one nucleus

*cells from slide **M***

*cells from slide **DW***

(ii) Describe **one** observable difference between the cells on slide **I** and the cells on slide **M**.

1. Cells on slide I are stained **yellow-brown/ orange** while cells on slide M are stained **blue**.

Accept AVP from candidate's drawings

[1]

- (b) In another study, a student observed cells found in regions **A** and **B** (Fig. 2.2), and carried out an unpaired t-test to compare the mean length of cells in both regions.

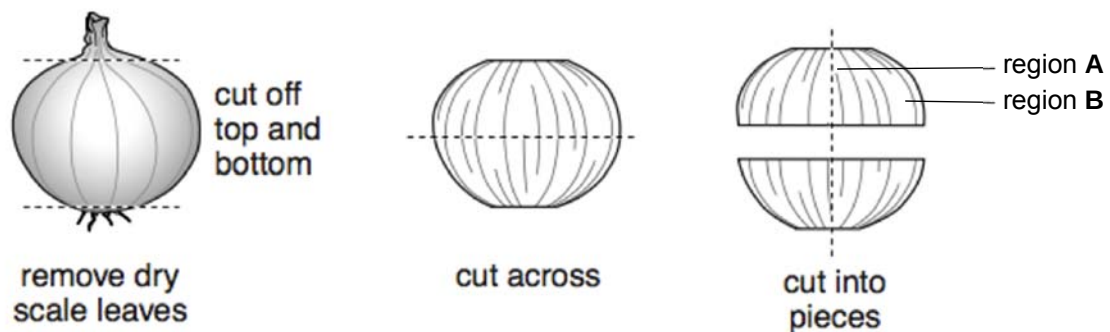


Fig. 2.2

Table 2.1 shows some of the student's results.

Table 2.1

Mean length of cells / μm		Calculated t-value
region A	region B	
280	292	1.789

The student conducted the t-test based on the 5% significance level, and referred to the following portion of the t-table (Table 2.2).

Table 2.2

	Significance level				
Degrees of freedom	20%	10%	5%	2%	1%
20	1.325	1.725	2.086	2.528	2.845

(i) State the total number of cells the student measured.

1. 22

[1]

(ii) Describe briefly how the student would have carried out the measurement of the cells in (i), in order to ensure the validity of his results.

1. ref. random selection of cells to be measured in regions A and B/ measured cells are from various areas in regions A and B.

[1]

(iii) Interpret the student's results.

1. ref. calculated / t value is smaller than the critical value;
2. At 5% significance level, difference between mean length of cells in regions A and B is not statistically significant/ any difference observed is due to chance.

[2]

(c) Fig. 2.3 is a simplified plan drawing of a section of the leaf of *Ammophila sp.*, a desert grass. In particular, the numerous folds that can be observed on the inner surface of this leaf are depicted.

The length of a fold is indicated by line L in Fig. 2.3.

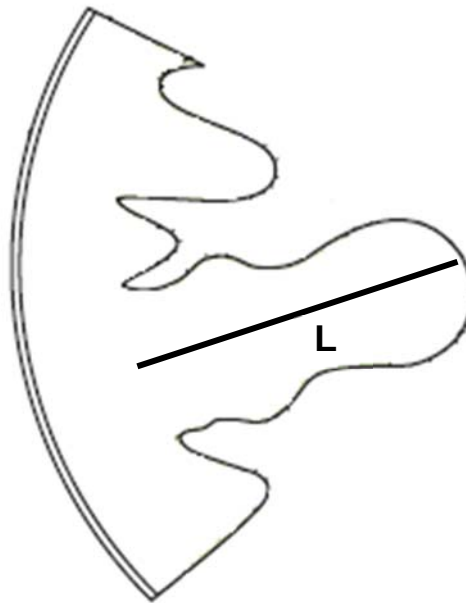


Fig. 2.3

Slide **S2** is a transverse section from a leaf of *Ammophila* sp.

Using the microscope, examine slide **S2**.

- (i) Locate the **largest** fold present in slide **S2**. Using the stage micrometer, determine its length. **Indicate this length in μm .**

You may lose marks if you do not show clear working or if you do not use appropriate units.

1. Indicate clearly objective lens used (*i.e.* 10x) + show calibration of stage micrometer with working (*i.e.* 1 eyepiece graticule division under x10 objective lens = 0.01 mm or 10 μm)
2. Indicate length of largest fold in eyepiece graticule units (*Accept: 40-70 units*)
3. Show working to determine actual length of largest fold
4. Indicate actual length in μm (1 mm = $10^3 \mu\text{m}$).

actual length μm [4]

- (ii) Describe **one** observable feature on the surface of the folds of the *Ammophila* leaf in Slide **S2**.

Explain how **this** feature may help the plant to grow in dry conditions by preventing water loss.

description of feature

1. Presence of many/ numerous hair-like projections

[1]

explanation

1. Ref. to trapping a layer of water vapour/ creating a region of high humidity in the air spaces, hence reducing transpiration.

[1]

[Total: 17]

Question 3

Climate change has led to noticeable changes in global temperatures. For example, average temperatures in the northern latitudes in the sub-tropics are now higher. In tropical countries such as Singapore, the traditionally cooler year-end months have also become increasingly warmer.

Such changes in global temperatures have impacts on the development of mosquito vectors such as *Aedes aegypti*, which is involved in dengue transmission.

The typical life cycle of *A. aegypti* is shown in Fig. 3.1 below.

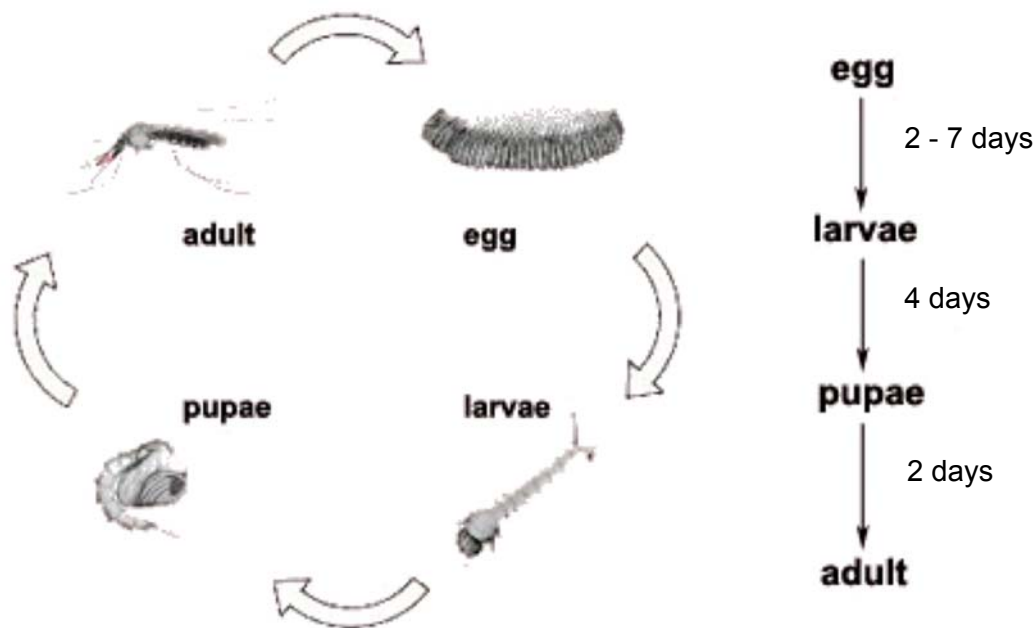


Fig. 3.1

Design an experiment to investigate the effect of temperature on the development of *A. aegypti*.

- (a) (i) With reference to Fig. 3.1, identify an appropriate period of the life cycle of *A. aegypti* to observe in this experiment, highlighting one significant advantage and disadvantage of observing this period.

- Egg → Larvae
- Advantage: 1. Morphologically dissimilar 2. Safe (no chance of biting)
- Disadvantage: 1. Time for this period varies widely (2 – 7 days) 2. Long period (up to 7 days)
- Larvae → Pupae
- Advantage: 1. Safe (no chance of biting)
- Disadvantage: 1. Both are morphologically similar looking/ there might be intermediates which are difficult to distinguish. 2. Long period (4 days)
- Pupae → Adult
- Advantage: 1. Morphologically dissimilar 2. Relatively shorter period
- Disadvantage: 1. Adult mosquitos are able to bite. 2. Difficult to determine start of pupae stage.

(ii) Predict the results of this experiment.

1. Duration of identified period would decrease until a certain/ optimum temperature, after which this duration would increase.

[1]

(b) Design your experiment accordingly to observe the period identified in (a) (i).

Your planning must be based on the assumption that you have been provided with the following equipment and materials which you **must** use:

- A sufficient number of *A. aegypti* at your starting stage of choice
- Fine-meshed netting and rubber bands
- 0.1% nutrient medium
- Thermal incubators
- Light microscope

You may select from the following materials and apparatus, and use appropriate additional apparatus:

- Normal laboratory glassware, e.g. test tubes, boiling tubes, petri dishes, measuring cylinders, glass rods, etc.
- Syringes
- Plastic pipettes
- Spatula
- Timer, e.g. stopwatch
- Distilled water

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it
- be illustrated by relevant diagrams, if necessary
- identify the independent and dependent variables
- describe the method with scientific reasoning used to decide the method so that the results are as accurate and repeatable as possible
- include layout of results tables and graphs with clear headings and labels
- use the correct technical and scientific terms
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[10]

[Total: 13]

MARKING POINTS

(marking points may be credited from diagrams etc)

1. Theory: Idea of how increase in temperature would cause an increase in enzyme activity/ metabolism, leading to faster development.
2. Independent variable*: temperature (°C) + give at least five sensible temperatures between 20 to 40 °C e.g. 20 °C, 25 °C, 30 °C, 35 °C, 40 °C.
3. Dependent variable*: average time taken to change from one identified stage to the next e.g. larva to pupa stage (h)
4. Controlled variable: Volume of 0.1% nutrient medium that eggs/ larvae/ pupae are placed in + give specified volume
5. Controlled variable: Frequency at which 0.1% nutrient medium is replaced + give specified interval
6. Controlled variable: Sample size/ number of organism used + specific number
7. Procedure – use of plastic pipette or spatula to transfer of eggs/ larvae/ pupae into nutrient medium in specified glassware
8. Procedure – choice of keeping identified stage in a petri dish (accept: boiling tube, beaker) + rationale e.g. to prevent overcrowding/ larger water surface area to obtain oxygen from surroundings
9. Procedure – checking that all samples are viable/ at the same stage at the start under the light microscope/ periodic examination of samples under light microscope to check development/ check for stage change
10. Procedure – use of fine-meshed netting and rubber band to cover petri-dish + rationale e.g. to prevent escape of adult mosquitoes
11. Procedure – keeping of samples in thermal incubators of different identified temperatures throughout experiment
12. Accuracy – Use of sufficiently large number, i.e. at least 10, of eggs/ larvae/ pupae to find average time taken to change from one identified stage to the next, hence minimising error.
13. Reproducibility – Repeat entire experiment twice with fresh *A. aegypti* samples, 0.1% nutrient medium to ensure reproducibility of data/ trend.
14. Recording – Appropriate results table with correct headings and units
15. Recording – Results graph of time taken to change from one stage to the next against temperature with the predicted trend
16. Safety* – Relevant risk and safety precaution discussed, e.g.
 - adult mosquitos are capable of biting; kill samples before adult stage by pouring samples onto dry surface/ at adult stage by freezing/ insecticide.
 - Thermal incubator/ microscope are electrical appliances – handle with dry hands to avoid electrocution.